One Sentence of Grounding: A Three-Configuration Hallucination Analysis of a Retrieval-Augmented Chatbot over Republic Act No. 9003

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***Abstract*—Statutory text demands exact answers, yet a general-purpose language model supplies a plausible figure with no signal that it is guessing. This paper documents the construction and evaluation of a retrieval-augmented generation (RAG) chatbot over the complete text of Republic Act No. 9003, the Philippine Ecological Solid Waste Management Act of 2000. Seven chapter files totalling roughly 108,000 characters were split into 239 chunks, embedded with all-MiniLM-L6-v2, indexed in ChromaDB, and queried through a Groq-hosted model under three prompt configurations. Deleting the scripted fallback sentence at temperature 1.0 did not induce hallucination; the model refused in its own wording. Deleting the single clause restricting it to retrieved context did, producing wholly ungrounded answers. A false refusal on an answerable query was traced to a chunk boundary severing a penalty from its offence. Prompt restriction, not temperature, governed grounding.**

Keywords—retrieval-augmented generation, hallucination, prompt engineering, vector databases, legal informatics, LLMOps

I. System Architecture

The system answers questions about Republic Act No. 9003, the Ecological Solid Waste Management Act of 2000 [1], which governs waste management for every local government unit in the Philippines. Its provisions are consulted by officials and citizens who need to know precisely what the law requires: which acts are prohibited, what fine attaches to each, which body holds which mandate. A general-purpose model handles this poorly. Its knowledge of any single statute is diffuse, and it gives no indication when it is guessing — a fabricated peso amount arrives in the same confident register as a correct one. Retrieval-augmented generation [2] addresses both halves by grounding each answer in retrieved text and permitting refusal.

The corpus is the complete statute, converted from a text-layer PDF into seven plain-text files, one per chapter. Plain text avoids any parsing layer that can fail, and splitting by chapter means the filename printed beside each retrieved chunk identifies which division of the law an answer came from — decisive in Section III-C.

Fig. 1 shows the pipeline. DirectoryLoader parses the files into Document objects; RecursiveCharacterTextSplitter divides them at paragraph, line and sentence boundaries into 239 chunks of at most 500 characters with 50 of overlap. Each chunk is embedded by all-MiniLM-L6-v2 [3], a distilled sentence transformer producing 384-dimensional vectors on CPU, and indexed in an in-memory ChromaDB collection. At query time the three nearest chunks are concatenated into the {context} placeholder of the system prompt and passed with the question to the model [4].

II. Implementation Details

Table I lists the parameters as actually executed.

TABLE I.
Pipeline Parameters as Executed

Parameter	Value
Source corpus	7 .txt files, RA 9003 (~108,000 chars)
Document loader	DirectoryLoader, glob **/*.*
Splitter	RecursiveCharacterTextSplitter
Chunk size / overlap	500 / 50 characters
Raw documents / chunks	7 / 239
Embedding model	all-MiniLM-L6-v2 (384-dim, CPU)
Vector store	ChromaDB, in-memory
Top-k retrieval	3 (k = 6 comparison also run)
LLM	openai/gpt-oss-20b via Groq
Temperature	0 (baseline); 1.0 (stress-test)

Values recorded from notebook output, not from the template specification.

Two deviations were forced by the environment. The specified model llama-3.1-8b-instant was decommissioned by Groq on 16 August 2026 [5], so results were produced on openai/gpt-oss-20b, its recommended replacement. LangChain 1.0 also moved the legacy chain builders into langchain-classic, so the template imports raised ModuleNotFoundError until migrated [6]. Both are LLMOps exposures rather than coding errors: a working pipeline broke because upstream dependencies changed on schedules outside the author's control. A model identifier hardcoded into a constructor is fragile and belongs in configuration.

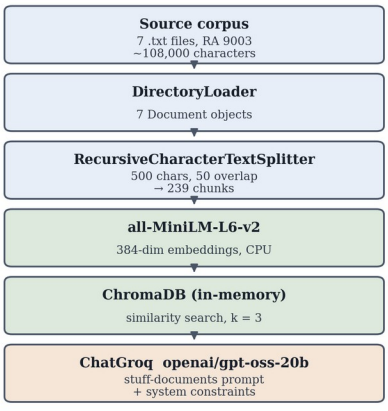


Fig. 1. Pipeline architecture. Blue stages run once at index time; green stages serve every query; the orange stage generates the answer.

III. Experimental Results and Hallucination Analysis

A. In-Domain and Out-of-Domain Behaviour

Table II reports the four evaluation queries.

TABLE II.
TEST QUERIES, EXPECTED RESPONSES, AND MODEL OUTPUTS

Query	Type	Expected	Actual output
Members of the National Solid Waste Management Commission	In-domain	14 government, 3 private sector	Correct; sourced to CHAPTER II.txt
Sources of the Solid Waste Management Fund	In-domain	Fines, permits, donations, GAA appropriations	Correct; sourced to CHAPTER V.txt
Penalty for littering in a public place	In-domain	₱300–₱1,000 or 1–15 days community service	False refusal (see Sec. III-C)
Average surface temperature of Venus	Out-of-domain	Fallback message	Exact fallback under Config A

The sources retrieved for the Venus query are more instructive than the refusal itself: quorum rules from Chapter II, the bill's passage dates from Chapter VII, penalty amounts from Chapter VI — three passages unrelated to planetary science. Nearest-neighbour search has no similarity threshold below which it declines; it yields k chunks regardless of distance. The retrieval stage cannot refuse, so every refusal observed here is a property of the prompt alone.

B. Baseline versus Ungrounded Configurations

Raising temperature from 0 to 1.0 while deleting the fallback sentence changes two variables together, so that comparison alone cannot attribute an outcome to either. A third configuration was added, holding temperature at 1.0 and additionally deleting the clause "Answer questions strictly using ONLY the provided context below." Each configuration was run three times on the identical query with an identical retriever.

TABLE III.
PROMPT CONFIGURATIONS AND OBSERVED OUTCOMES

Config	Restriction	Fallback	Temp.	Outcome on Venus query
A	present	present	0	Exact scripted fallback message
B	present	deleted	1.0	Refused all 3 trials, own wording
C	deleted	deleted	1.0	Answered from training data, all 3 trials

Config B refused on all three trials. This is a genuine result rather than a failed experiment: deleting the fallback sentence removes the scripted wording for declining, not the permission to decline, because the restriction clause survives it. The model duly refused in its own phrasing ("the information you're looking for isn't included in the context that was provided") rather than the prescribed sentence.

Config C answered in full, reporting an average surface temperature near 735 K and attributing it to a runaway greenhouse effect, while three chunks of Philippine waste-management law sat unused in the context window. Temperature was identical to Config B, so it cannot account for the difference; the restriction clause can. Temperature governs how the model says things, the prompt governs what it is permitted to say.

Two observations sharpen this. The Config C answers are factually correct, so this is not fabrication but ungroundedness — the more insidious failure, since nothing marks the moment the system stopped being a document-based assistant, and the sources printed alongside would lend a wrong answer unearned credibility. Config C also varied across trials in substance rather than wording: 735 K, then a hedged 460–470 °C, then 464 °C. Temperature's effect migrates from phrasing into facts precisely when the model draws on parametric memory instead of a fixed context.

C. A False Refusal and Its Cause

The littering query, which Section 49(a) answers directly, was refused despite CHAPTER VI.txt being retrieved. Chunking had split Section 49(a) mid-sentence: the chunk holding the amounts begins "hundred pesos (P300.00) but not more than One thousand pesos..." with no indication of which offence it governs. Section 48 names littering; Section 49 sets the fine and refers back only by cross-reference, so a natural-language question retrieves the offence but not the penalty. Raising k to 6 did not help, confirming the fault is semantic rather than one of coverage, while querying in the statute's own wording retrieved the orphaned chunk as top hit, confirming it was reachable. Cross-references carry almost no semantic signal, making legal text hostile to fixed-size chunking and locating the fix at the chunking stage.

This is the cost side of strict grounding: a refusal caused by retrieval failure and one caused by a genuinely absent answer are indistinguishable to the user. A system tuned only against hallucination will decline questions it could have answered.

IV. CONCLUSION

Three safeguards were exercised, and the experiments give each a measurable weight. Prompt restriction proved decisive: the comparison between Configs B and C shows a single clause of plain English was the entire difference between a grounded assistant and an ungrounded one. That is a thin safeguard for how much it carries, and it argues for treating system prompts as versioned, tested artefacts rather than casually edited text.

Retrieval-based grounding contributed what the prompt could not. Printing source chunks beside every answer is what made Section III-C diagnosable; without provenance there is no way to separate a generation failure from a retrieval failure, and the two demand opposite remedies. Retrieval alone is not a safeguard, as the Venus query shows.

API-key protection was the cheapest of the three and the easiest to neglect. Reading the key through getpass into an environment variable kept it out of the submitted notebook, where a hardcoded literal would have travelled with the file into any repository. The three safeguards operate at different layers — what the model may say, what evidence it is given, and who may invoke it — and none substitutes for another. Future work should target chunking: a structure-aware splitter treating each Section as an atomic unit would likely resolve the failure that raising k could not.

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